



Lecture #6: Mixed Strategies

Mitchell Scott

Math 190: Mathematics of Sustainability, Emory University

February 1, 2026



EMORY
UNIVERSITY



Table of Contents

1 Mixed-strategies

Nash Equilibrium with Mixed Strategies

IESDS and Rationalizable Equilibria with mixed strategies

Summary





Introduction to Mixed-strategies

Recall the game:

		Player 2	
		L	R
Player 1	U	3,0	0,0
	M	0,0	3,0
	D	1,0	1,0

This means that sometimes it would be better to randomize your strategies.



Rock Paper Scissors

Let's call this strategy $\sigma = (\sigma(R), \sigma(P), \sigma(S)) = (\frac{1}{2}, \frac{1}{2}, 0)$. As long as $\sigma(R), \sigma(P), \sigma(S) \geq 0, \sigma(R) + \sigma(P) + \sigma(S) = 1$, it is a valid mixed-strategy profile. But how do figure out the expected payoff of this kind.

Let Player 2 play $\sigma_2 = (\frac{1}{2}, \frac{1}{2}, 0)$. What should player 1 do?

This means that $BR_1(\sigma_2) = \{P\}$.





Matching Pennies

Let's play a game. Both you and your friend have pennies, and you get to decide which side you want up. If you both play heads or both play tails, player 1 gets both pennies. If the two pennies don't match, then player 2 gets both pennies. What should we do?





Matching Pennies (cont.)





Matching Pennies (cont.)

Now for the best response if Player 1 chooses T .

This means we want to maximize the payoff, so if $v_1(H, \sigma_2) > v_1(T, \sigma_2)$, then we will choose H . This happens when $q > \frac{1}{2}$ since $v_1(H) > 0 > v_1(T)$.

We can also see that if $q < \frac{1}{2}$ then $v_1(T) > v_1(H)$. When $q = \frac{1}{2}$, $v_1(H) = v_1(T)$.

Now let's look at the best response for Player 2 responding to σ_1 .





Matching Pennies (cont.)

$$v_2(\sigma_1, H) = \sigma_1(H)v_2(H, H) + \sigma_1(T)v_2(T, H)$$

and

so their best response profile will be





Matching Pennies (cont.)





Understanding Mixed Strategies

Remark

We notice that $v_1(H, \sigma_2^) = v_1(T, \sigma_2^*)$ and $v_2(\sigma_1^*, H) = v_2(\sigma_1^*, T)$. This is not a coincidence.*

Remark

A pure strategy is a special case of a mixed strategy, where $\sigma(s) = 1$ for one s , a pure strategy.





Understanding Mixed Strategies (cont.)

Remark

A player who plays a mixed strategy in a Nash Equilibrium doesn't do that for the (direct) benefit, the payoff of all pure strategies that they are mixing between are all the same. Instead they do this so that their opponent doesn't take advantage of the deviation from the Nash equilibrium. In other words, Player 1 plays mixed strategy such that Player 2 is indifferent between the pure strategies in support of the optimal mixed strategy σ_2^ .*





Combining Pure and Mixed Nash Equilibria

Theorem

The total number of mixed-strategy Nash equilibria is odd or infinite.

		Player 2	
		<i>C</i>	<i>R</i>
Player 1	<i>M</i>	0, 0	3, 5
	<i>D</i>	4, 4	0, 3

We can see that:

This means that the pure strategy Nash equilibria are





Combining Pure and Mixed Nash Equilibria (cont.)

To find the mixed strategy Nash equilibrium. Let's say that it is (σ_1^*, σ_2^*) is a Nash equilibrium, where σ_i^* is the mixed Nash equilibrium for player i . This means that $v_1(M, \sigma_2^*) = v_1(D, \sigma_2^*)$ and $v_2(\sigma_1^*, C) = v_2(\sigma_1^*, R)$.

To find this we need to say that in Player 2's Nash equilibrium, they will pick C q percent of the time, which means they will pick R $(1 - q)$ percent of the time.





Combining Pure and Mixed Nash Equilibria (cont.)

Since we know the Nash Equilibrium is happening when $v_1(M, \sigma_2^*) = v_1(D, \sigma_2^*)$, we set these two equations equal to each other, to see





Combining Pure and Mixed Nash Equilibria (cont.)

Now we turn our attention to what Player 2 should do with Player 1's strategy $\sigma_1^* = (\sigma_1^*(M), \sigma_1^*(D)) = (p, 1 - p)$.

Putting this all together, we see that





Left, Right, Center, Revisited

		Player 2		
		L	C	R
Player 1	U	5, 1	1, 4	1, 0
	M	3, 2	0, 0	3, 5
	D	4, 3	4, 4	0, 3

We are going to try to find an IESDS strategy. Consider Player 2's mixed strategy $\sigma_2 = (\sigma_2(L), \sigma_2(C), \sigma_2(R)) = (0, \frac{1}{2}, \frac{1}{2})$

We can verify that this strictly dominates L



Left, Right, Center, Revisited (cont.)

This means we eliminate L , as it is strictly dominated which means it is never a best response.

		Player 2	
		C	R
Player 1	U	1, 4	1, 0
	M	0, 0	3, 5
	D	4, 4	0, 3

We see that no pure strategy dominates another pure strategy from either player.

Let's check if there is a mixed strategy that we can find.

Consider if Player 1 mixes between M and D so

$$\sigma_1 = (\sigma_1(U), \sigma_1(M), \sigma_1(D)) = \left(0, \frac{1}{2}, \frac{1}{2}\right)$$

.





Left, Right, Center, Revisited (cont.)

This means we can find the expected payoff for this mixing strategies.

This tells us to eliminate U since it is dominated by σ_1 . This leads us to

		Player 2	
		C	R
Player 1	M	0, 0	3, 5
	D	4, 4	0, 3





All Equilibrium

Remark (IESDS Equilibrium)

IESDS strategies

Remark (Rationalizable Equilibrium)

Rationalizable strategies

Remark (Nash Equilibrium)

At the risk of being repetitive, let me emphasize the requirements for a Nash equilibrium:

- 1 *Each player is playing a best response to his beliefs.*
- 2 *The beliefs of the players about their opponents are correct*

All Equilibrium (cont.)

Remark (Pareto Optimal)

A Pareto optimal solution is when the strategy is not Pareto Dominated by another solution. It is at least as good as the other strategy and strictly better in at least one way.

Theorem (Nash's Existence Theorem)

Any n -player normal-form game with finite strategy sets S_i for all players has a (Nash) equilibrium in mixed strategies.

